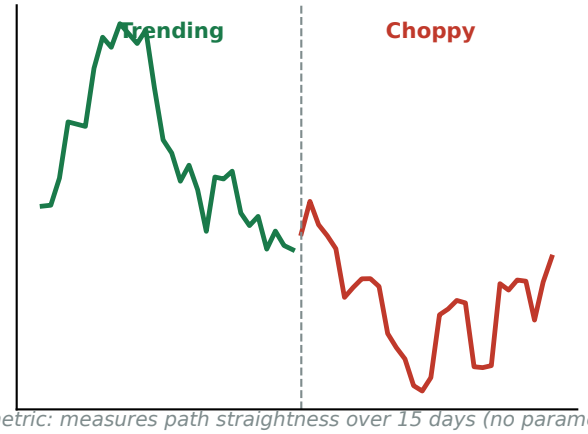
Regime Ensemble

Detecting market conditions to reduce risk — not just maximise returns

SPY · 2000–2025 · v3.1

Two Orthogonal Detectors — One Ensemble



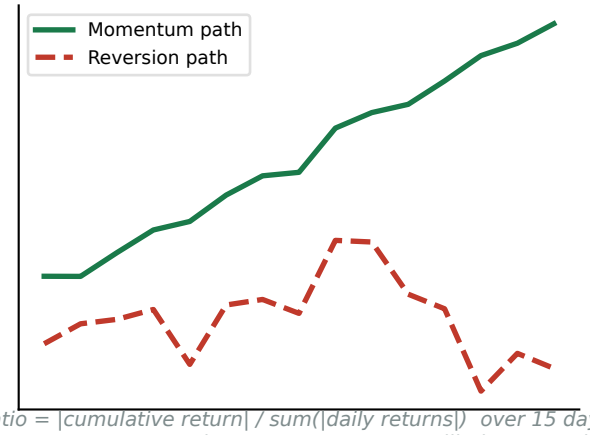
Geometric: measures path straightness over 15 days (no parameters).
Markov: estimates a hidden 3-state process from the full return history.
Combined via simple mean — no weight fitting, no in-sample optimisation.

The Key Finding — Mixed Regime

Regime	Mean ret/day	T-stat (p)
Momentum	+0.018%	1.31 (0.189)
Mixed	+0.108%	3.21 (0.001)**
Reversion	-0.012%	-0.35 (0.729)

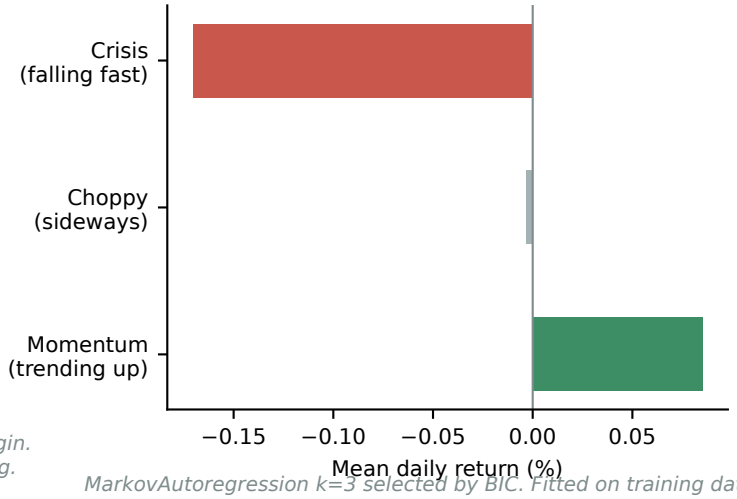
Mixed = detectors disagree. Despite uncertainty, SPY drifts upward. We hold a half-position (+0.5), not cash. This is the most significant signal.

Detector 1 — Geometric (path shape)



ratio = |cumulative return| / sum(|daily returns|) over 15 days.
Ratio near 1.0 = monotonic move. Near 0.0 = oscillation around origin.
Thresholds are adaptive percentiles (70th/30th) -- zero hand-tuning.

Detector 2 — Markov AR(1), k=3



MarkovAutoregression k=3 selected by BIC. Fitted on training data only; Hamilton filter applied to test data (no look-ahead). Crisis regime (P>0.50) suppresses all buy signals regardless of geometric signal.

Headline Results (SPY 2000-2025, zero transaction costs)

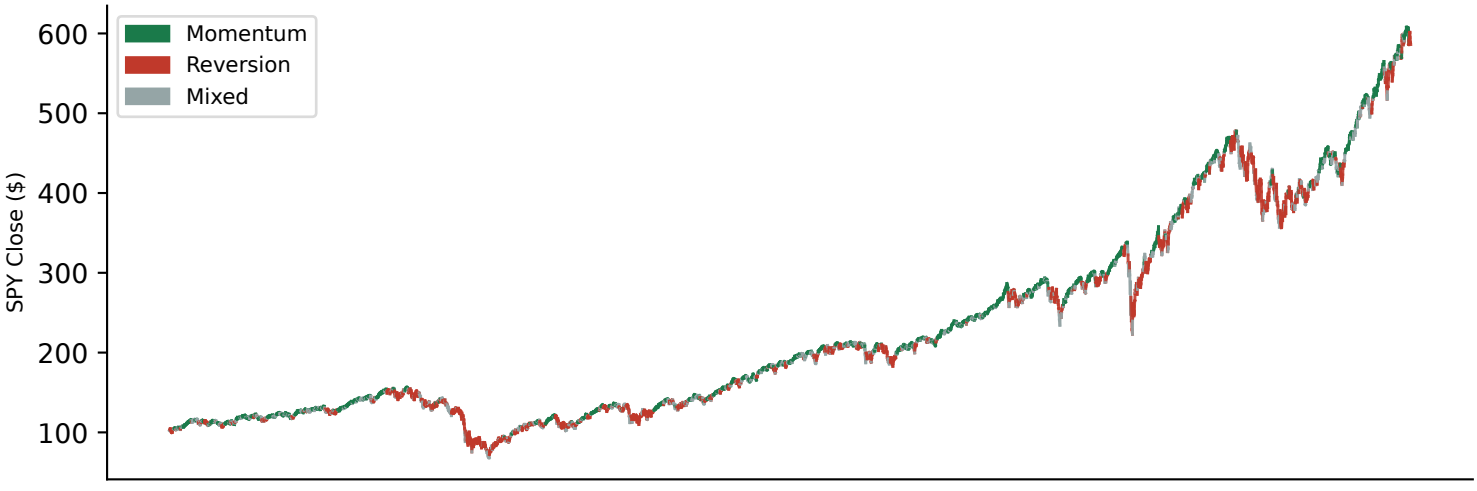
	CAGR	Sharpe Ratio	Max Drawdown	T-stat (p-val.)
Strategy (Regime Ensemble)	**+5.7%**	**0.68**	**-16.7%**	**3.12 (0.002)**
Buy & Hold (benchmark)	+8.6%	0.44	-56.5%	2.02 (0.043)

⚠ At 10bps round-trip costs, Sharpe drops to 0.28. Strategy breaks even around 20bps. Transaction costs are material — see page 3.

Models in Action — SPY 2000-2025 (Real Data)

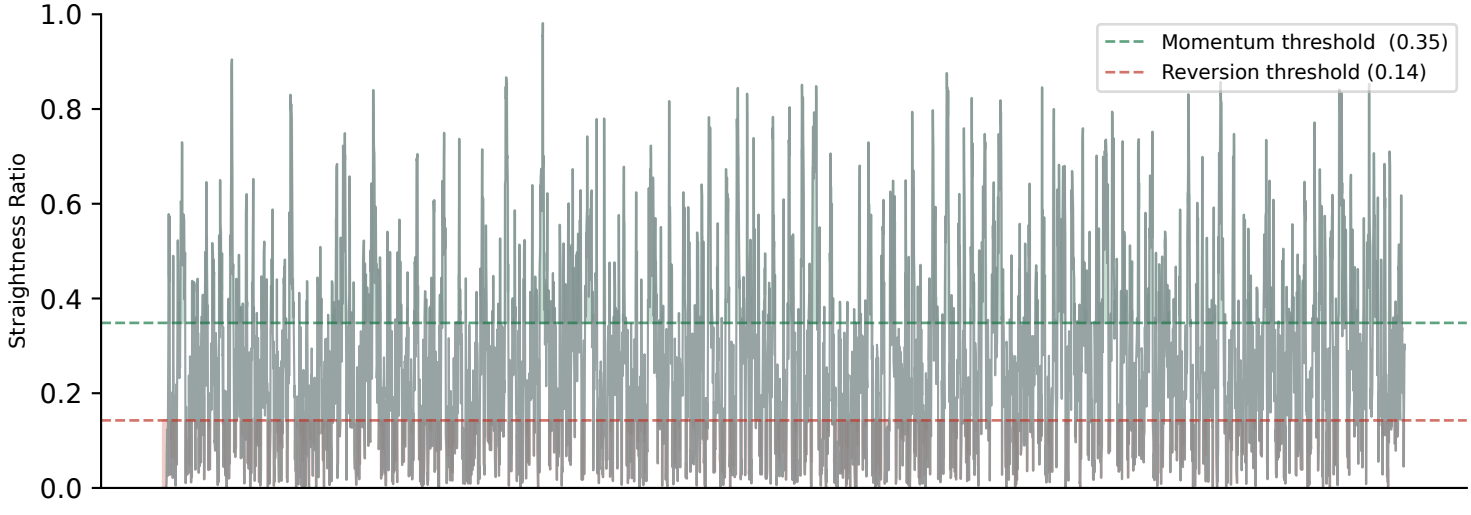
Three panels: price coloured by regime, geometric detector signal, Markov probabilities

SPY Price — Coloured by Ensemble Regime



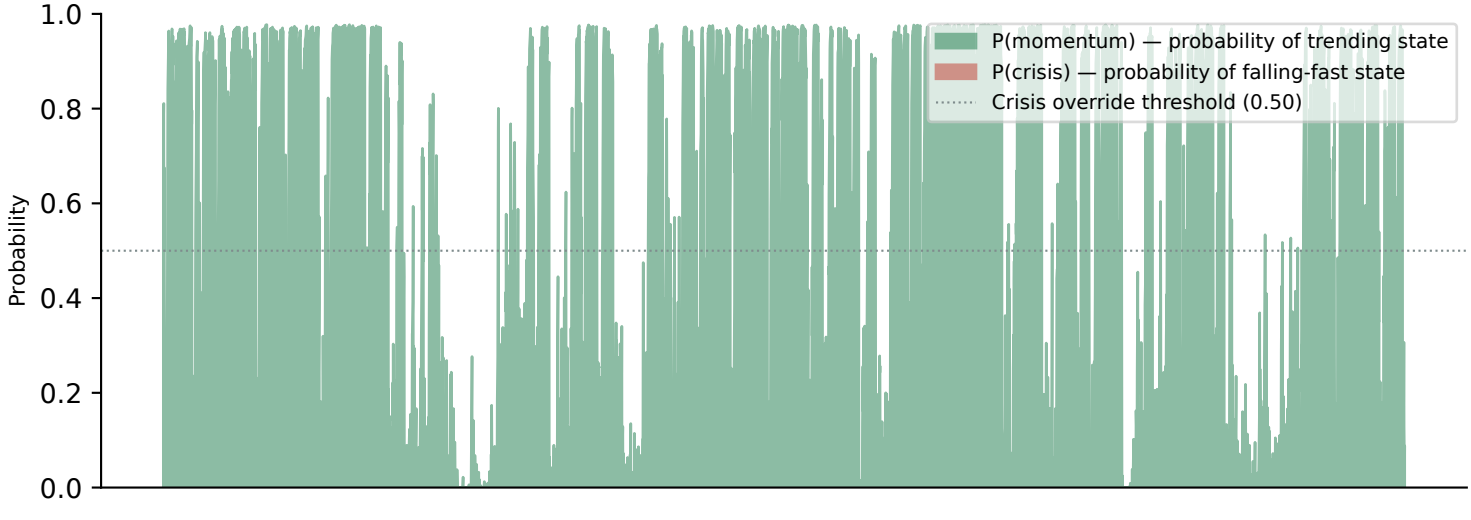
Green = momentum (+1). Grey = mixed (+0.5, half long). Red = reversion/crisis (cash).
25 years shown: dot-com crash (2000-02), GFC (2008), COVID (2020), bear market (2022).

Geometric Detector — Straightness Ratio (15-day window)



Near 1.0 = price moved in a straight line (momentum). Near 0.0 = price zigzagged (mean-reversion).
Thresholds are adaptive percentiles — no fixed values to hand-tune.

Markov Detector — Filtered State Probabilities (k=3 regimes)

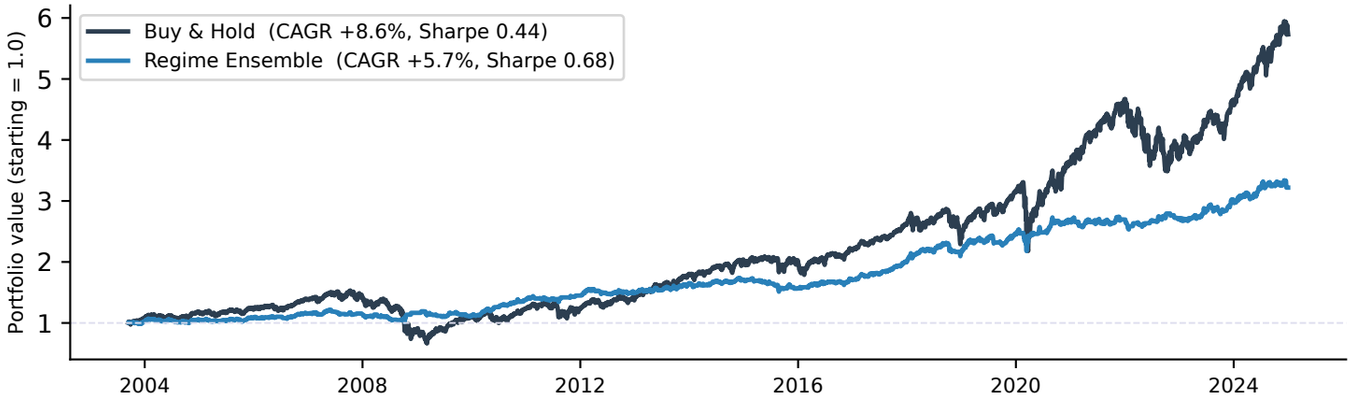


When P(crisis) crosses 0.50, buy signals are suppressed regardless of other indicators.
Filtered probabilities only — no look-ahead bias.

Results & Limitations

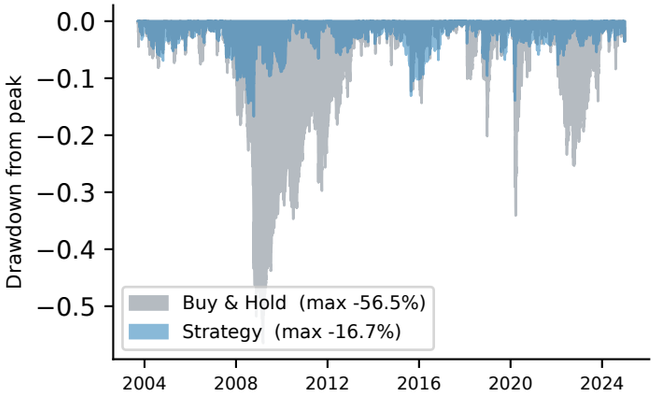
Equity curves, drawdowns, transaction cost sensitivity, and what to watch out for

Equity Curves — Strategy vs Buy & Hold (0 bps costs)



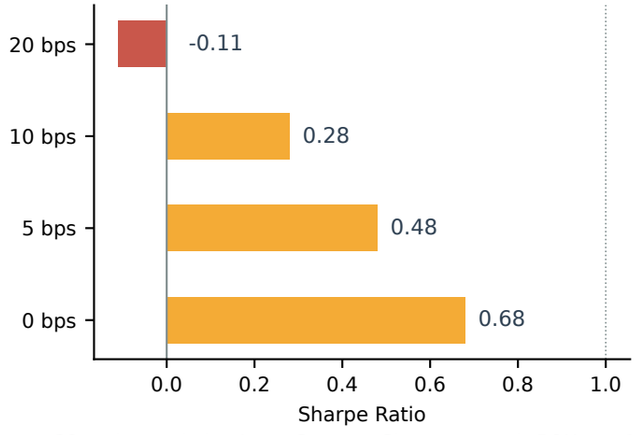
Strategy Sharpe (0.68) beats buy-and-hold (0.44) with far lower drawdown (-16.7% vs -56.5%).
CAGR trails B&H (+5.7% vs +8.6%) -- the edge is risk-adjusted return, not raw return.

Drawdown Comparison



Strategy reduces drawdowns across all major crashes (dot-com, GFC, COVID, 2022).

Transaction Cost Sensitivity



~20 regime switches/year. Strategy Sharpe exceeds B&H to ~15bps;
breaks even at ~20bps. Use --min-hold 3 to cut switches and extend the range.

Key Limitations — Read Before Drawing Conclusions

⚠ **Single asset, no out-of-sample generalisation test**
25 years of SPY. Regime structure is not validated on other tickers, sectors, or international markets.

⚠ **Transaction costs constrain the edge**
~20 switches/year. Sharpe >B&H up to ~15bps round-trip. Use --min-hold 3 to extend profitability to ~20bps.

⚠ **In-sample threshold calibration**
Percentile thresholds (70th/30th) and ensemble cutoffs are fit on the full dataset. A real deployment requires expanding-window recalibration from a fixed start date.

⚠ **Reversion signal is not significant**
Individual regime T-stats: mixed=3.21** (p=0.001), momentum=1.33 (p=0.18), reversion=-0.35 (p=0.73). Do not short on the reversion signal.

⚠ **Strategy CAGR trails buy-and-hold**
+5.7% vs +8.6% over 25 years at zero cost. The edge is Sharpe (0.68 vs 0.44) and drawdown (-16.7% vs -56.5%), not raw return.